

Demand Forecasting Using Random Forest and Artificial Neural Network for Supply Chain Management

Problem statement

Description

Supply chain management (SCM) is considered as one of the key elements to leveraging any company's success. Demand forecasting is one of the main issues of supply chains. It aimed to optimize stocks, reduce costs, and increase sales, profit, and customer loyalty. One challenge of modeling retail data is the need to make decisions based on limited history. If Christmas comes but once a year, so does the chance to see how strategic decisions impacted the bottom line.

Without proper demand forecasting processes in place, it can be nearly impossible to have the right amount of stock on hand at any given time. Any service must deal with a lot of perishable raw materials which makes it all the more important for such a company to accurately forecast daily and weekly demand. Too much inventory in the warehouse means more risk of wastage, and not enough could lead to out-of-stocks — and push customers to seek solutions from your competitors. In this challenge, get a taste of demand forecasting challenge using a real dataset.

Industry Alignment

1. Supply chain Industries
2. Retailer, Consumer, Manufacturing, Services, Automobile etc.

Challenges

We have been loose in the definition and included problems that were a time series before obfuscation or have a clear temporal component. Hence, we will solve 3 challenges. They are –

1. **Company asks:** How many quantities of various products will be sold in next month? Or in any given time.
2. **Company says:** For various products I don't know how much quantity should I put in my purchase order before ordering to my supplier?
3. **Company asks:** Will my product "Product X" will be sold or not? Is this product now obsolete in the market?

Proposed solution

We shall tell, for various products how much to procure / stock in the shop / warehouse for asked timeline, as simple as that. We give a summary of base forecasting techniques such as time series and regression methods, support vector regression model, feature reduction approach, deep learning methodology, and a new final decision integration strategy. In our proposed system, nine different time series algorithms including moving average (MA), exponential triple smoothing (ETS), Holt-Winters, Croston, ARIMA methods, and three different Regression Models are employed.

Keywords: Machine learning, Demand and sales forecasting, Supply chain analytics, Supply Chain Management, Traditional forecasting methods.

Dataset

Recent trending news in India –

1. Indian Auto Industry Growth Story About to Collapse: Tata Motors MD
2. All major automobile manufacturers report sharp decline in August 2019 sales
3. Auto sector crisis: August 2019 turns out to be another gloomy month in India

Let's consider automobile spare parts dataset. Below are the attributes required for the use case –

1. Product ID
2. Product Name
3. Product Category
4. Invoice Date
5. Invoice Number
6. Shipped / Sold Quantities
7. Sale Type
8. Price per unit
9. Customer type
10. Customer number
11. Stock location
12. Hub location

ROI Realization time

Why: before you embark on an expensive, time-consuming trajectory, you want to investigate the feasibility of realizing a data science solution based on your data, but you currently lack the necessary skills and expertise.

What: we realize a prototype data science solution for your business goals using representative sample data.

How:

1. **Feature engineering:** we perform feature engineering (incl. feature construction and selection)
2. **Reference data solution:** we define a methodology and identify and select relevant analysis and modelling algorithms we evaluate the results of applying it on a reference dataset.
3. **Visualization dashboard:** we provide prototype dashboards that visually represent the analysis results with respect to the business goals

Deliverables:

1. **Feature engineering report:** a detailed technical document containing a description of the selected features, an explanation of the methodology and an overview of the achieved results
2. **Proof-of-concept implementation:** a reference prototype implementation demonstrating.
 - a. The analysis and modelling methodology that you can use on similar (future) data, using open source tools.

b. Several data analytics dashboards with advanced visualizations

Phase wise duration

Phase	Activities	Duration
Project kickoff	Requirement Gathering: The Supply Chain Scientist presents the different phases of implementation and the various deliverables that can be expected by the client.	2 weeks
Data specifications	The Supply Chain Scientist produces the data specifications required for the implementation of the project. These specifications are reviewed and validated together with the client. In particular, this document shall define the data to be extracted from the client's IT systems.	3 weeks
Knowing Data	After validating the specifications, the first set of observations are discussed. Explorations are done, anecdotal evidences reports are published. To address the limitations of anecdotes, statistical approach is taken - 1. Collect Data 2. Describe Statistically on how to visualize. 3. Exploratory Analysis: Look for patterns, differences, and other features that address the questions we are interested in. At the same time we will check for inconsistencies and identify limitations. 4. Estimation: We will use data from a sample to estimate characteristics. 5. Hypothesis Testing: Where we see apparent effects, like a difference between two groups, we will evaluate whether the effect might have happened by chance.	4 weeks

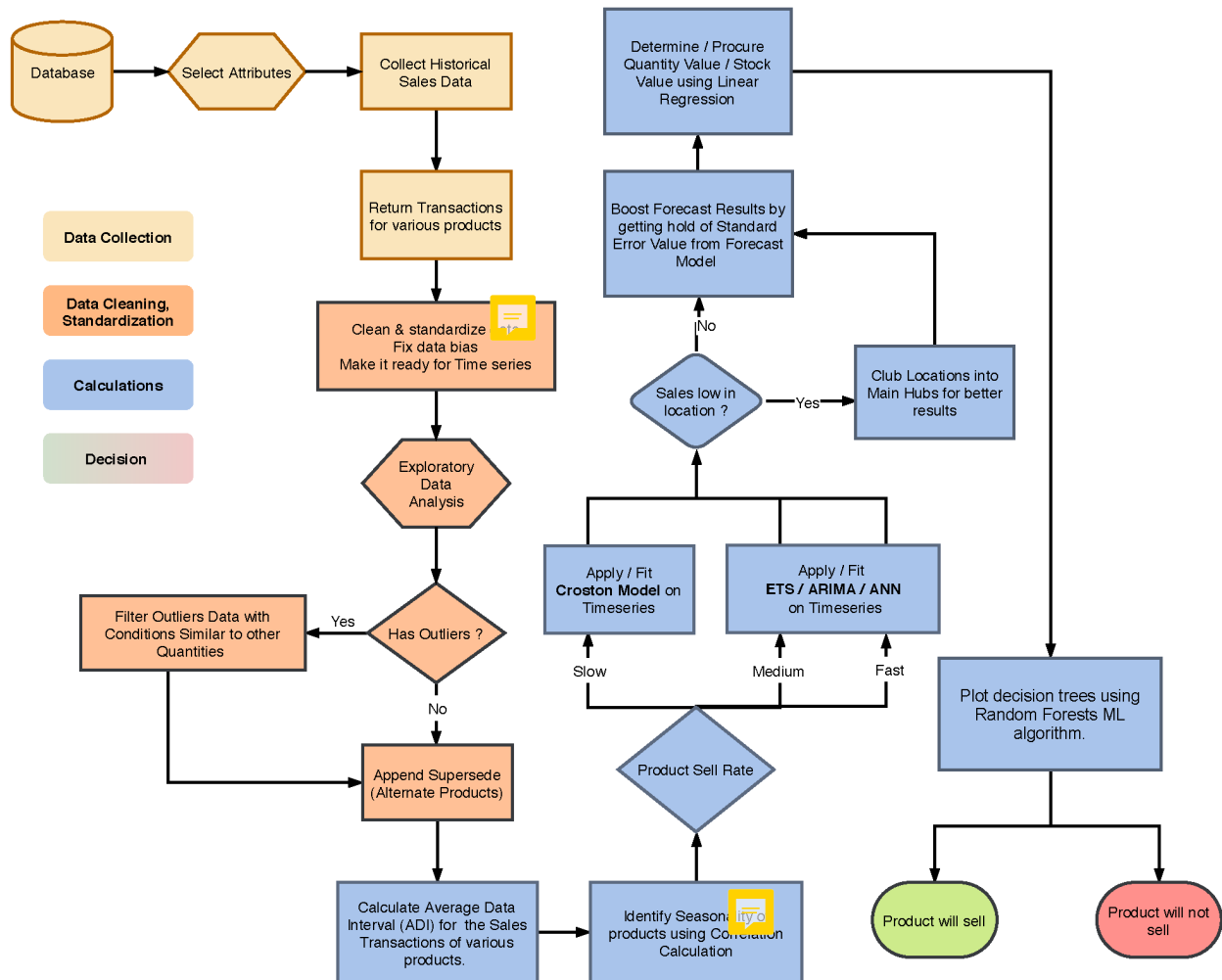
Validating the data	The Supply Chain Scientist performs an in-depth investigation of the client's dataset content. In particular, sanity checks are introduced to monitor the quality of the data according to multiple metrics. The goal is to make sure that - 1) the dataset is properly refreshed by the upload process, 2) the dataset correctly reflects the reality of the business and 3) the dataset is coherent and complete enough for supply chain optimization purposes. In terms of deliverables, during this phase the Supply Chain Scientist provides the client with various dashboards that assess the health of the data. These dashboards can be used by the client even for purposes that go beyond the Quantitative Supply Chain initiative itself - as part of their internal data quality assurance process for example.	2 weeks
Mid-project audit	10 weeks following the initial kickoff, a meeting is set up to evaluate the project completion status. The objective of this "audit" is to address, as early as possible, the problems that may be experienced during the implementation, especially those that could delay the production phase. This "audit" consists of an exchange between the Supply Chain Scientist and the client, based on a checklist that is communicated to the client in advance by the Supply Chain Scientist, right after the kick-off meeting.	2 weeks
Implement and Upload automation	Once both parties validate the overall quality of the dataset that has been uploaded so far, the client implements an automated process that transfers their dataset to us Deloitte on a regular basis - ideally daily. At the same time, on Deloitte's side, the data health logic - with its multiple checks - is scheduled to be refreshed after every upload. The models are executed on data and results are derived using accelerators	4 weeks

Setting up the optimization	From this point on, the Supply Chain Scientist has all the necessary ingredients for implementing the optimization of decisions which have been agreed on with the client previously. Therefore, he implements scripts to generate different quantitative outputs: operational purchase suggestions, dispatch suggestions, etc. The figures produced by these scripts can be visualized in dashboard form. At this stage, these dashboards represent only a preliminary version of the final dashboards and need to be revised together with the client.	2 weeks
Feedback & fine-tuning	The client's requests to make some kind of alteration or "tweak" the different outputs usually lead to some fine-tuning of the scripts written by the Supply Chain Scientist. There are many parameters and methods that can be adopted to adequately align the characteristics of the supply chain being optimized with the optimization logic. When the methodology itself is of strategic importance to the client, this is explicitly discussed between the client and the Supply Chain Scientist.	2 weeks
Production	After several rounds of fine-tuning and revision, the client reaches a stage when he comes to trust the logic implemented by the Supply Chain Scientist. At this point, the client can start using the service in production, that is, he can directly execute the supply chain decisions as originally computed by the software. When the client validates that the solution is production-ready, the Supply Chain Scientist delivers a documentation which ensures the maintainability of the solution.	2 weeks

Support & maintenance	The solution is operational and is used by the client while the Supply Chain Scientist monitors the smooth daily execution of the data pipeline. Calls are regularly organized between the client and the Supply Chain Scientist to verify that the optimization delivers the expected degree of supply chain performance. Moreover, supply chains are not static, thus, business or IT changes, small or big, need to be reviewed: a new warehouse, shift of the market, new process, etc. The Supply Chain Scientist proposes suitable modifications to accommodate these different changes. Checkpoint calls are scheduled with an agreed-upon frequency, typically monthly.	Until Warranty period agreed between client and Deloitte
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High level Design flow Architecture

Overview



The following section outlines the design flow of the proposed solution. Using the different colors used in the flow diagram we have segregated into 4 major phases.

Phase 1

1. Data collection
2. Exploratory Data Analysis
3. Hypothesis testing

Phase 2

1. Extracting required attributes
2. Data cleaning and standardization
3. Outliers detection

Phase 3

1. Determine Time series
2. Location clubbing

3. Apply / Fit Traditional Forecast models
4. Forecast Sales
5. Forecast Demand

Phase 4

Use Random Decision Forest to answer question – If Product “X” will be sold or not? / If Product “X” is now obsolete?

Methodologies to be used

Exploratory & Descriptive Data analysis

It answers questions about what is happening or what has happened by providing real time information. It is an excellent source of real-time information that can be applied in the different business units of a company. For instance, having information about the exact location of the goods in a chain will allow a SC manager to know whether warehouses are under or over-utilized.

Traditional Time series forecasting

In our proposed system, nine different time series algorithms including moving average (MA), exponential triple smoothing (ETS), Holt-Winters, ARIMA, Croston methods, and three different Regression Models are employed. In time series forecasting models, the classical approach is to collect historical data, analyze these data underlying feature, and utilize the model to predict the future

Table 1. shows algorithm definitions and parameters which are used in the proposed system. These algorithms are commonly used forecasting algorithms in time series demand forecasting domain.

Table 1: Time series algorithms used in demand forecasting.

Models	Formulation	Definition of Variables
Regression Model 1	$y_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \beta_3 X_{3,t} + \beta_4 X_{4,t} + \epsilon_t$	Variability in weekly customer demand; special days and holidays (X1), discount dates (X2), days when the product is closed for sale (X3), and sales (X4) that cannot explain these three activities but are exceptional.
Regression Model 2	$y_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \beta_3 X_{3,t} + \sum_{j=1}^{52} \gamma_j W_{j,t} + \epsilon_t$	R1 is modeled by adding weekly partial-regressive terms ($W_{j,t}$; $j = 1, \dots, 52$).
Regression Model 3	$y_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \beta_3 X_{3,t} + \sum_{j=1}^{12} \alpha_j M_{j,t} + \sum_{j=1}^4 \gamma_j \overline{W}_{j,t} + \epsilon_t$	The R3 model adds the shadow regression terms to the R1 model for each month of the year $M_{j,t}$, $j = 1, \dots, 12$, and for the months of the week $\overline{W}_{j,t}$, $j = 1, \dots, 4$ end result.
SVR (Support Vector Regression)	$y_t = \text{SVR}(\text{input variables})$	The SVR prediction is used for forecasting y_t by using inputs X_1, X_2, X_3 mentioned regression model 1, each month of the year $M_{j,t}$, $j = 1, \dots, 12$, and the months of the week $\overline{W}_{j,t}$, $j = 1, \dots, 4$.
Exponential Smoothing Model	$y_t = \alpha \hat{y}_{t-1} + (1 - \alpha) y_{t-1} + \epsilon_t$	Exponential smoothing is generally used for products where there is no trend or seasonality in the model.
Holt-Trend Methods	$\begin{aligned} \hat{y}_t &= L_t + T_t \\ L_t &= \alpha y_t + (1 - \alpha)(L_t - 1) \\ T_t &= \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \end{aligned}$	Additive Holt-Trend model has been evaluated for products with level (L) and trend (T).
Holt-Winters Seasonal Models	$\begin{aligned} \hat{y}_t &= L_t + T_t + S_{t+p+1} \\ L_t &= \alpha y_t + (1 - \alpha)(L_t + T_t) \\ T_t &= \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \\ S_{t+p+1} &= \gamma \left(\frac{D_{t+1}}{L_{t+1}} \right) + (1 - \gamma)S_{t+1} \end{aligned}$	Additive Holt-Winters Seasonal models have been evaluated for products with level (L), seasonality (S) and trends (T)
Two-Level Model 1	$\begin{aligned} y1_t &= \alpha \hat{y}_{t-1} + (1 - \alpha) y_{t-1} + \epsilon_t \\ y2_t &= \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \beta_3 X_{3,t} + \beta_4 X_{4,t} + \epsilon_t \\ y_t &= y1_t - y2_t + \epsilon_t \end{aligned}$	In this method, R1 model was first applied to the time series. Later, the demand values estimated by this model are subtracted from the customer demand data and the residual values are fitted with exponential correction and Holt-Winters Exponential Smoothing Model. Estimates of these two metrics were collected and the product demand estimate was calculated and evaluated.
Two-Level Model 2	$\begin{aligned} y1_t &= L_t + T_t \\ L_t &= \alpha y1_t + (1 - \alpha)(L_t - 1) \\ T_t &= \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \\ y2_t &= \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \beta_3 X_{3,t} + \beta_4 X_{4,t} + \epsilon_t \\ y_t &= y1_t - y2_t + \epsilon_t \end{aligned}$	In this method, R1 model was first applied to the time series. Later, the demand values estimated by this model are subtracted from the customer demand data and the residual values are fitted with exponential correction and Holt-Winters Trend Model. Estimates of these two metrics were collected and the product demand estimate was calculated and evaluated.
Two-Level Model 3	$\begin{aligned} y1_t &= L_t + T_t + S_{t+p+1} \\ L_t &= \alpha y_t + (1 - \alpha)(L_t + T_t) \\ T_t &= \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \\ S_{t+p+1} &= \gamma \left(\frac{D_{t+1}}{L_{t+1}} \right) + (1 - \gamma)S_{t+1} \\ y2_t &= \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \beta_3 X_{3,t} + \beta_4 X_{4,t} + \epsilon_t \\ y_t &= y1_t - y2_t + \epsilon_t \end{aligned}$	In this method, R1 model was first applied to the time series. Later, the demand values estimated by this model are subtracted from the customer demand data and the residual values are fitted with exponential correction and Holt-Winters Seasonal Model. Estimates of these two metrics were collected and the product demand estimate was calculated and evaluated.

ARIMA and ANN models

In this study, it is intended to use time series and artificial neural networks as modelling and forecasting tools. There are several studies, which have used these tools in an independent, comparative or combined manner in search of greater accuracy. In recent publications, the comparison of different models for time series forecasting can be observed, involving the ARIMA and ANNs models. In most cases, the ANNs

models exceeded the forecasting quality over the ARIMA models. Nevertheless, there are also cases in which the ARIMA models worked better [35], and the importance of selecting an appropriate ANNs architecture to solve the corresponding problem is shown. As an alternative, it has been proposed to combine models deploying different methods to improve the time series forecasting, commonly obtaining better results than in isolated models. Some of these hybrid models have been directly applied to the problem of forecasting spare-parts demand in hybrid industries, combining linear regression and ANNs and working with small sample sizes. We propose the hybrid models used for this study. The ARIMA and ANNs models have been compared, thus obtaining mixed conclusions in terms performance superiority when forecasting, and therefore proposes a hybrid methodology combining both models in order to take the advantages of one another in linear and nonlinear modelling. The results obtained by Hybrid models were better in the case of mixed models, than those obtained with each model individually. Several authors have relied on Hybrid models or have developed models based on the same principle of hybridization.

Implemented Models – The resulting models for each part are first presented, in order to later analyse the results in the sample and post-sample periods. The selected model for each part is shown in

Table 1. The sample period spans up to the 38th month and the post-sample period spans up to the 48th month.

Table 1. Architecture for the selected models to each part being studied

Part	Classic Models	ANN			Hybrid		
		ARIMA	Delays	Neurons	ARIMA	Delays	Neurons
A	Average	ARMA(3,4)	3	7	ARMA(3,4)	2	5
B	Moving averages, 2 delays	ARMA(5,4)	2	8	ARMA(5,4)	2	5
C	Moving averages, 3 delays	ARMA(5,4)	3	7	ARMA(5,4)	3	8

ETS

Exponential smoothing is a time series forecasting method for univariate data. This can be used for fast pace selling products, or the one having highest average sale interval. Time series methods like the Box-Jenkins ARIMA family of methods develop a model where the prediction is a weighted linear sum of recent past observations or lags. Exponential smoothing forecasting methods are similar in that a prediction is a weighted sum of past observations, but the model explicitly uses an exponentially decreasing weight for past observations. Specifically, past observations are weighted with a geometrically decreasing ratio. Triple exponential smoothing applies exponential smoothing three times, which is commonly used when there are three high frequency signals to be removed from a time series under study. There are different types of seasonality: 'multiplicative' and 'additive' in nature, much like addition and multiplication are basic operations in mathematics.

Croston

This is specially for slow pace selling products, or the one having less average sale interval.

A new technique to forecast products with intermittent demand. The idea could be summarized in three simple steps:

1. Evaluate the average demand level when there is a demand occurrence.
2. Evaluate the average time between two demand occurrences.
3. Forecast the demand as the demand level (when there is an occurrence) multiplied by the probability to have an occurrence.

The Croston model answers the question How much demand will we have on average per period?

Model

Now that we understand the basic idea of the Croston model, we need to answer two questions:

1. How do we estimate the probability to have a demand occurrence?
2. How do we estimate the demand level when there is an occurrence?

Croston used a technique close to the one used by Holt & Winters to answer these questions. Like for the different exponential smoothing models, we will be looking at both the previous estimation of each variable (i.e. level & periodicity) and its most recent observation.

Demand level

Let's note our level estimate a (like for the exponential models) and note the actual demand observations d . We will update our level estimate a only when we have an actual observation, so that as in the different exponential smoothing models, we use a learning parameter α ($0 < \alpha < 1$) to allocate more or less importance to the most recent observations or to the historical ones.

Periodicity

Let's note the estimation of the time between two demand occurrences p (for periodicity), and the time elapsed since the previous demand occurrence q . We will only update p when we have a demand occurrence.

Note that we again use α as the learning parameter. This is the same parameter we used to estimate the demand level.

Forecast

The forecast is very simple: it is the demand level (a) divided by the periodicity (p).

Summary

We can then summarize the whole model. As you can see, when there is no demand observation, none of the parameters are updated. This will be one of the limitations of this model.

Insights

Croston vs Simple Exponential Smoothing

The biggest addition of the Croston method compared to the simple exponential smoothing is its ability to estimate the time between two demand occurrences. Let's see on the figure below if this helps us to forecast intermittent products.

Linear Regression

Demand forecasters begin a regression analysis by identifying the factors or drivers of demand.

In this method the estimation of demand is done through the past data available as well as through the factors influencing the demand. We have seen how the demand for the product is affected by several factors. The demand function is stated with respect to the factors influencing demand. Let's see what the steps are involved in regression method.

Variables detection: The first and foremost task is to recognize the variables that would impact the product for which demand forecasting has been undertaken.

Past data collection: Once the factors are determined the next step is to gather the data. The past data with respect to the above variables should be gathered.

Selection of Demand function: After variables selection and data collection the forecaster then must determine the demand function. Demand for the product (sales of the product) which is to be computed is the dependent variable. The factors influencing the demand are the independent variables.

Function Estimation: The estimation of the demand function would be the next step in the regression analysis. The coefficient values are found out in this step. The value of coefficient of determination is arrived at.

Forecast Derivation: The final step in the regression analysis is to derive the forecast on the basis of the predicted values of coefficients. In other words, the regression method estimates the values of incomes, price, prices of related products and promotional expenditure for the coming times. On the basis of these values the future demand (sales) for the product is estimated.

Decision Trees: Random Forests / Random Decision Forests

Random forest is an ensemble machine learning algorithm that is used for classification and regression problems. Random forest applies the technique of bagging (bootstrap aggregating) to decision tree learners. There are many reasons why random forest is so popular. Reasons are:

1. Ensemble learning prevents overfitting of data
2. Bootstrapping enables random forest to work well on relatively small datasets
3. Predictors can be trained in parallel
4. Decision tree learning enables automatic feature selection

A suitable algorithm to classify automobile parts detected by a multiple inductive loop system, developed for measuring sales & demand parameters in a heterogeneous and no-lane disciplined traffic. The proposed classification scheme employs Random Forest (RF) algorithm. This scheme is suited not only for classifying the detected obsolete products but also for counting them accurately under a mixed sales & demand condition. The algorithm has been implemented and tested. Its performance has also been compared with other algorithms based on threshold values and signature patterns. The threshold, signature and RF based algorithms use the number of loops a vehicle occupies as an important factor for classification. Results from a prototype system developed show that the RF based algorithm provides better accuracy compared to the threshold based and signature-based methods.

Viability and Effort

Effort Estimates

The below section outlines the effort estimates for end-to-end implementation and quality assurance.

Activity	Duration	
Explore & Study	500	Man hours
Development & Unit Testing	400	Man Hours
UI Development / Reporting	300	Man Hours
Quality Assurance: Technical & Functional	300	Man Hours

Tech stack – Basic / Primary

Visualization: Tableau

You can connect with various types of data and convert into easy comprehensible Tableau Reporting, Tableau data visualization and Tableau dashboard. It's best for data exploration & describe data statistically

Python with Data manipulation, Mathematics Libraries

Python perform tasks hassle-free during prototype building. Python also enables developers to roll out programs and get prototypes running, making the development process much faster. Once a project is on its way to becoming an analytical tool or application

Excel / Any spreadsheet

Unix Environment

Tech stack – After upgrading

Apache Hadoop on Unix Environment

Get the data out in the first place to do analysis and Hadoop is the technology that stores large volumes of data, which we can work on.

Python on Apache Spark with Data manipulation, Mathematics Libraries

Spend more time wrangling data than making models. Traditional tools like Pandas provide a very powerful data manipulation toolset. Transitioning to big data tools like PySpark allows one to work with much larger datasets but can come at the cost of productivity.

Data ingestion and warehousing tools: Sqoop and Hive

Scheduler: Oozie / Autosys

Visualization: Tableau

Build interactive slideshows and presentations of your data in minutes. Use dashboards as a front end for Python model code, allowing viewers to intuitively interact with Statistical & Machine Learning models.

- ## Market Place

- ## Appendix

```
val finalresults =  
new PlotDecisionTreesUsingRandomForests(  
new EligiblePreferredLocationMutate(  
new MinMaxMutate(  
new FixForecastValue( //Boosting Forecast value  
new SafetyStockMutate(  
new ForecastMutate(  
new PrepareForecastingData(  
new ForecastPeriodLeadTimeMutate(  
new BuyingLocationMutate(  
new ADICalculation(  
new HighTransactionsValidRecords( // Remove Outlier Transactions  
new NetSaleValidProducts( // Remove all products from transactions dataframe which have NetSale Zero  
new LocationClubbing(  
new SupersedeProducts( //Append Supersede Products to Transactional Data  
new LossSalesWithCondition(  
new SalesReturnExportWithCondition(  
new SalesReturnWithoutExport(  
new PlainData())))))))))))
```

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